

chap_strate_CONSTRUCTION

2025-07-10

```
library(sampling)
library(survey)

## Loading required package: grid
## Loading required package: Matrix
## Loading required package: survival
##
## Attaching package: 'survival'
## The following objects are masked from 'package:sampling':
##
##      cluster, strata
##
## Attaching package: 'survey'
## The following object is masked from 'package:graphics':
##
##      dotchart
data(rec99); attach(rec99)

arrondi <- function(a)
{
  o <- order(a-floor(a),decreasing=T) #ordonne selon parties décimales
  n <- ceiling(sum(a)) #calcule la taille totale de l'échantillon
  #avec prise en compte des arrondis éventuels au niveau des nh
  nbar = n-sum(floor(a)) #calcule le nombre de nh
  #à arrondir à l'unité par excès
  nouv <- floor(a) # arrondit tous les nh à l'unité par défaut
  if (nbar!=0){nouv[o[1:nbar]] <- nouv[o[1:nbar]] + 1}
  #arrondit certains nh par excès si nécessaire
   #(les nh avec les plus faibles écarts à l'entier supérieur)
  if (any(nouv == 0))
  {print("Attention, une strate est à tirage nul")}
  nouv}

```

Construction des strates

Principales méthodes pour le choix de bornes

```
STRATQUART <- ifelse(LOG < quantile(LOG,0.25), 1,
                     ifelse(LOG < quantile(LOG,0.5), 2,
                              ifelse(LOG < quantile(LOG,0.75), 3, 4)))

STRATALEA <- c(rep(1,139),rep(2,139),rep(3,138),rep(4,138))

```

Mise en œuvre avec le package stratification

```
library(stratification)

(cum=strata.cumrootf(LOG, n = 70, Ls=4, nclass=60))

## Given arguments:
## x = LOG
## nclass = 60, n = 70, Ls = 4
## allocation: q1 = 0.5, q2 = 0, q3 = 0.5
## model = none
##
## Strata information:
##      |      type rh |      bh    E(Y)    Var(Y)  Nh nh  fh
## stratum 1 | take-some  1 |  161.27   77.33   1522.61 308 13 0.04
## stratum 2 | take-some  1 |  534.43  295.95  10671.04 146 16 0.11
## stratum 3 | take-some  1 | 1430.03  850.58  53044.60  67 16 0.24
## stratum 4 | take-some  1 | 4491.00 2221.15 529513.52  33 25 0.76
## Total                                     554 70 0.13
##
## Total sample size: 70
## Anticipated population mean: 356.1625
## Anticipated CV: 0.03214144
cum$bh #les bornes des strates

## [1] 161.2667 534.4333 1430.0333
#[1] 161.2667 534.4333 1430.0333
cum$Nh #les tailles des strates

## [1] 308 146 67 33
#[1] 308 146 67 33
cum$nh #les tailles des échantillons par strate

## [1] 13 16 16 25
#[1] 13 16 16 25

geo <- strata.geo(LOG, n = 70, Ls=4)
geo$bh

## [1] 52.77734 232.12066 1020.89263
#[1] 52.77734 232.12066 1020.89263
geo$Nh

## [1] 100 265 138 51
#[1] 100 265 138 51
geo$nh

## [1] 1 11 24 34
#[1] 1 11 24 34

LH <- strata.LH(LOG, n = 70, Ls=4,
  algo="Kozak", algo.control=list(trymany=FALSE))
LH$bh
```

```
## [1] 181.5 533.5 1302.5
```

```
#[1] 181.5 533.5 1302.5
```

```
LH$Nh
```

```
## [1] 326 128 64 36
```

```
#[1] 326 128 64 36
```

```
LH$nh
```

```
## [1] 15 13 14 28
```

```
#[1] 15 13 14 28
```

```
bh <- strata.bh(LOG, bh=cum$bh, n = 70, Ls=4, takeall=1)
```

```
bh$Nh
```

```
## [1] 308 146 67 33
```

```
#[1] 308 146 67 33
```

```
bh$nh
```

```
## [1] 11 13 13 33
```

```
#[1] 11 13 13 33
```

```
bh <- strata.bh(LOG, bh=geo$bh, n = 70, Ls=3, takenone=1)
```

```
bh$Nh
```

```
## [1] 100 265 138 51
```

```
#[1] 100 265 138 51
```

```
bh$nh
```

```
## [1] 0 11 25 34
```

```
#[1] 0 11 25 34
```

```
geo$nh
```

```
## [1] 1 11 24 34
```

```
#[1] 1 11 24 34
```

```
head(LH$stratumID)
```

```
## [1] 1 1 1 3 1 1
```

```
## Levels: 1 2 3 4
```

```
#[1] 1 1 1 3 1 1
```

```
#Levels: 1 2 3 4
```

```
head(geo$stratumID)
```

```
## [1] 2 2 2 3 2 2
```

```
## Levels: 1 2 3 4
```

```
#[1] 2 2 2 3 2 2
```

```
#Levels: 1 2 3 4
```

Comparaison des différentes méthodes de stratification par étude de Monte Carlo

```
N <- nrow(rec99)
```

```
n <- 70
```

```

#(n.optlog <- arrondi(n*Nh*sdlog/sum(Nh*sdlog)))

STRATCUM <- cum$stratumID
STRATGEO <- geo$stratumID
STRATLH <- LH$stratumID
rec99 <- data.frame(rec99,STRATQUART,STRATALEA,STRATCUM,STRATGEO,STRATLH)

CVsimu <- function(STRA)
{
  Nh <- table(rec99[[STRA]])
  rec99.order <- rec99[order(rec99[[STRA]]), ]
  sdlog <- tapply(LOG,rec99[[STRA]],sd)
  n.optlog <- arrondi(n*Nh*sdlog/sum(Nh*sdlog))

  if (n.optlog[1] == 1) {
    n.optlog[1] <- 2
    n.optlog[4] <- n.optlog[4] - 1
  }

  set.seed(123456)
  nsim <- 10000
  est <- numeric(nsim)
  for (i in 1:nsim)
  {
    stsi <- sampling::strata(rec99.order,STRA,size=n.optlog,
method="srswor")
    stsi.data <- getdata(rec99.order,stsi)
    stratum_formula <- as.formula(paste0("~", STRA))
    ech.stsi <- svydesign(id=~1,strata=stratum_formula,
weights=rep(Nh/n.optlog,n.optlog),
fpc=rep(n.optlog/Nh,n.optlog),
data=stsi.data)
    est[i] <- svymean(~LOGVAC, ech.stsi)[1]
  }
  cv <- sd(est)/mean(est); print(cv)
}

CVsimu("STRATALEA")

## [1] 0.1909887
CVsimu("STRATQUART")

## [1] 0.09028185
CVsimu("STRATLOG")

## [1] 0.07129638
CVsimu("STRATCUM")

## [1] 0.06921321
CVsimu("STRATGEO")

## [1] 0.07591606

```

```
CVsimu("STRATLH")
```

```
## [1] 0.069921
```

```
# STRATALEA CV : 19.2%  
# STRATQUART CV : 8.9 %  
# STRATLOG CV : 7.1 %  
# cum CV : 7.1 %  
# geo CV : 7.6 %  
# LH CV : 6.9 %
```